

Key Points:

- A novel BC data set was developed using machine learning and linked to socio-demographic data to assess exposure disparities in India
- Estimated high all-cause, CVD, and respiratory mortality risks due to BC, with rural exposure driven by biomass and seasonal burning
- Cutting BC emissions can lower PM_{2.5} by 15% and yield greater health benefits per unit reduction

Supporting Information:

Supporting Information may be found in the online version of this article.

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Black Carbon-Attributable Mortality and Its Disparity Across the Population Subgroups in India

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Abstract Black carbon (BC) is an essential short-lived climate pollutant and a critical component of health-damaging fine particulate matter (PM_{2.5}), posing significant risks to public health and the climate system. Despite its dual impact, comprehensive assessments of population-level BC exposure, its associated mortality burden, and its disparities across population subgroups remain limited in India. We used an advanced machine learning algorithm to estimate annual BC mass concentration at a 1-km × 1-km spatial scale from 2016 to 2021. Following the global burden of disease (GBD) framework, we integrated the meta-analysis-derived risk estimates with demographic and epidemiologic attributes and estimated BC-attributable all-cause, cardiovascular disease, and respiratory mortality (per 100,000 population). Further, utilizing the socio-demographic information from the National Family Health Survey, we assessed the disparity in BC exposure and attributable burden across sub-populations. Annual population-weighted BC exposure varied from 0.4 to 13.7 µg/m³ in India between 2016 and 2021. While BC exposure remained stagnant over the years, its disparity across most demographic subgroups has diminished in recent years. We estimated annual BC-attributable all-cause, cardiovascular disease, and respiratory mortality (per 100,000 population) to be 133 (95% uncertainty intervals: 113–155), 7 (5–9), and 9 (5–14), respectively. BC-attributable mortality was found to be higher among females, other backward classes, and economically deprived subgroups than their demographic counterparts. Our results demonstrate that prioritizing BC emission reduction would improve ambient air quality, result in a larger health benefit (for every unit reduction in concentration), and slow down regional warming, thereby creating a win-win situation for India.

Plain Language Summary Black carbon (BC) is a harmful air pollutant that affects both human health and the climate. In India, where air pollution is extremely high, we still know little about how BC exposure impacts different social subgroups and contributes to early deaths. In this study, we used machine learning to estimate BC pollution across India from 2016 to 2021 at a fine spatial scale. We combined these estimates with health data to calculate how many deaths were linked to BC exposure, especially from heart and lung diseases. We also looked at how BC exposure varies across income, gender, and social subgroups. Our findings show that while BC levels remained mostly stable over time, vulnerable sub-populations such as women, poorer communities, and socially disadvantaged subgroups continue to face greater health risks. On average, BC exposure contributed to 133 deaths (per 100,000 population) per year in India. Reducing BC pollution would not only save lives but also help reduce the pace of climate change. These findings highlight the importance of targeting BC emissions as a public health and environmental priority.

1. Introduction

Black carbon (BC), a major constituent of fine particulate matter (PM_{2.5}) and a critical short-lived climate pollutant (SLCP) emitted from the incomplete combustion of fossil fuels and biomass, plays a key role in modulating Earth's radiation balance, degrading air quality, and driving adverse health outcomes (Venkataraman et al., 2016; Wei et al., 2023). Although India is recognized as one of the world's largest BC emission hotspots (Bond et al., 2013), the national ground-based BC monitoring network remains sparse and spatially fragmented, and a nationwide, long-term BC exposure database at high spatio-temporal resolution has yet to be established. Consequently, BC exposure-related studies in India have been constrained geographically and temporally. Most investigations have focused on climate impacts using data from a limited number of continuous monitoring stations or short-term field campaigns (Blanco-Donado et al., 2022; Kompalli et al., 2021; Liu et al., 2020), or

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have relied primarily on chemical transport modeling frameworks (Budhavant et al., 2023; Ghosh et al., 2020; Meehl et al., 2012). On the other hand, as compared to the rapidly expanding global literature on BC-attributable health impacts (Hang et al., 2023; Nichols et al., 2013; Song et al., 2022; Zhao et al., 2024; Zhu et al., 2023), epidemiological evidences from India remains extremely limited. To our knowledge, only two studies till date have examined the acute effects of BC exposure on all-cause mortality in two Indian cities, namely Delhi and Varanasi (Joshi et al., 2022; Singh et al., 2021); while two others have estimated BC-related mortality across the Indo-Gangetic Plain (IGP) region (Pathak et al., 2024; Verma et al., 2022). The lack of BC exposure data sets at high spatio-temporal resolution has constrained regional and sub-national air quality management and climate policy planning, and has limited the generation of consistent, indigenous epidemiological evidence on BC-related health impacts. Advancements in high-resolution BC exposure data sets and associated modeling methodologies would facilitate systematic monitoring of BC accumulation across emission hotspots over extended periods, thereby enabling more effective, epidemiologically informed air quality management within the framework of India's National Clean Air Program (NCAP).

A substantial body of environmental justice (EJ) research from the United States and elsewhere have demonstrated pronounced heterogeneity in ambient $PM_{2.5}$ exposure across demographic sub-populations (Clark et al., 2017; Rosofsky et al., 2018), prompting the integration of equity considerations into clean air policy design (de Souza et al., 2023). However, these studies have largely focused on disparities in ambient $PM_{2.5}$ concentration, while inequalities in exposure to its chemical constituents, such as BC, remain poorly characterized in the global literature. In our previous work (Sarkar et al., 2026), we demonstrated disparity in ambient $PM_{2.5}$ exposure across demographic subgroups in India; yet, subgroup-level disparities in exposure to BC, a major and toxic component of $PM_{2.5}$, remain unexplored in the Indian context. India's pronounced social, economic, and geographic diversity implies that individuals within the same demographic subgroup may experience considerable differences in BC exposure levels depending on demographic distributions, proximity to emission sources, and local socioeconomic conditions (Kerr et al., 2024). Therefore, assessing urban-rural disparities in BC exposure across states and demographic subgroups, and examining their temporal evolution, is critical for informing targeted, equity-focused policies aimed at advancing EJ and tracking India's progress toward national clean air targets.

Emerging evidence from high-income countries indicate that demographic subgroups that are exposed to different levels of air pollution also bear disproportionate air pollution-related acute and chronic health burdens (Geldsetzer et al., 2024; Kerr et al., 2024). While the Global Burden of Disease (GBD) framework has firmly established associations between ambient $PM_{2.5}$ exposure and adverse health outcomes (Murray, 2024), health burden assessments attributable to specific $PM_{2.5}$ constituents, including BC, remain absent. Furthermore, the GBD framework possesses two critical limitations. First, it relies on age- and sex-specific exposure-response functions to estimate relative risks (RRs), derived predominantly from cohort studies conducted in low-exposure settings across developed countries (Burnett & Cohen, 2020), whereas epidemiological evidence from high-exposure regions of the Global South remains sparse and geographically limited, resulting in the underrepresentation of large, low-income sub-populations (Worsley-Tonks et al., 2022). Secondly, as long-term air pollution exposure in India is highly heterogeneous across gender, wealth-index, household cooking fuel types, caste-based subgroups (analogous to ethnicity in Western contexts) (Sarkar et al., 2026); therefore, considering homogeneity in air pollution attributable health burden across these subgroups limits the identification of vulnerable ones for prioritized policy implementations to achieve socio-environmental equities. Nonetheless, existing EJ-based studies to estimate subgroup-specific health burden have applied uniform RR estimates across all sub-populations (Geldsetzer et al., 2024; Kerr et al., 2024), thereby overlooking critical differences in exposure distributions, baseline disease rates, and epidemiological susceptibility. Such simplifications may obscure meaningful disparities and bias health burden assessments. Addressing these critical knowledge gaps require the development of a comprehensive, equity-oriented framework that explicitly incorporates socio-demographic heterogeneity by estimating subgroup-specific relative risks, allocating *population-at-risk* and baseline disease rates accordingly, an approach yet to be systematically implemented in a socially diverse country like India.

Here, we leveraged onto recent advancements in machine learning (ML) methods to predict BC mass fraction (f_{BC}) from 2016 to 2021 which has been trained using daily scale CTM outputs for 2015 across India and other meteorological, geospatial, and demographic proxies. This predicted f_{BC} was integrated with daily scale satellite-derived ambient $PM_{2.5}$ concentration to estimate BC concentration at finer spatial resolution (1-km $\times$ 1-km), which were subsequently averaged for annual statistics; and the performance of our retrievals was evaluated

against the in situ observations (see Section 2). Further, we integrated BC concentration with the demographic information from the 4th (2015–16) and 5th (2019–21) rounds of the National Family Health Survey (NFHS, [n.d.](#)) to estimate urban-rural disparities in BC exposure across demographic subgroups, and its changes over the time points. Lastly, we relied on recent meta-analysis derived RR estimates to compute all-cause, cardiovascular disease (CVD), and respiratory health impacts attributable to BC exposure across the sub-populations, integrating their differential demographic and epidemiological attributes (see Section 2). Our results provide a comprehensive BC exposure data set for further epidemiological studies and an opportunity for policymakers to use our data and framework to track progress in controlling anthropogenic BC emissions, estimate potential health benefits and alleviate environmental inequalities in the foreseeable future.

2. Methods

2.1. Chemical Transport Model (CTM)

The Weather Research and Forecasting model coupled with Chemistry (WRF-Chem version 3.9.1) was utilized to simulate atmospheric conditions and aerosol properties across India, providing essential data for our BC exposure modeling. The model was configured at a horizontal resolution of $27\text{-km} \times 27\text{-km}$ and 48 vertical layers from the surface to 50 hPa for the year 2015, covering all major regions of the Indian subcontinent (Figure S1 in Supporting Information S1). The model integrated atmospheric physics and chemistry to simulate $\text{PM}_{2.5}$ concentrations and other aerosol parameters, including BC, on a daily scale (Agarwal et al., 2024). The MADE/SORGAM module (Modal Aerosol Dynamics Model for Europe/Secondary Organic Aerosol Model) was used for secondary inorganic and organic aerosols (Schell et al., 2001), employing a three-modal scheme of aerosol size distribution with internally mixed particles in each model (Zhao et al., 2010). Meteorological fields were assimilated every 12 hr using the three-dimensional variational assimilation technique, improving alignment with observations and enhancing surface $\text{PM}_{2.5}$ predictions. Assimilation data sets were sourced from the National Center for Environmental Prediction (rda.ucar.edu/data_sets/ds337.0/), incorporating diverse observational inputs like surface reports and satellite data. Anthropogenic emissions were modeled by combining SMOG-India for regional emissions (Sadavarte & Venkataraman, 2014) with the global Community Emissions Data System to capture diverse sources (Hoesly et al., 2018). The BC mass fraction (f_{BC}) was calculated for each grid cell by dividing the CTM model-simulated BC mass concentration by the total ambient $\text{PM}_{2.5}$ concentration, depicted in Figure S2 of Supporting Information S1.

2.2. Satellite-Derived Data Products

We utilized a range of satellite data sets for our ML modeling framework: NO_2 and O_3 from the Ozone Monitoring Instrument (OMI) onboard Aqua, and aerosol optical depth (AOD). The OMI level-3 data provide tropospheric column densities of criteria pollutants at a uniform resolution of $0.25^\circ \times 0.25^\circ$. These pre-processed data sets were incorporated directly into the model without further spatial interpolation. We used long-term satellite-derived annual mean ambient $\text{PM}_{2.5}$ concentrations averaged over 7 years (2015–2021) to represent air pollution exposure, given the substantial spatial gaps in India's ground-based monitoring network (Brauer et al., 2019). High-resolution ambient $\text{PM}_{2.5}$ concentrations across India were estimated using the methodology developed in our earlier work (Katoch et al., 2023), which addressed limitations in satellite-based air quality assessments arising from missing aerosol optical depth (AOD) retrievals. These retrieval gaps were particularly prominent during the monsoon and post-monsoon seasons and over complex terrain, mainly due to cloud cover and high surface reflectance affecting the Moderate Resolution Imaging Spectroradiometer (MODIS)-AOD product retrieved using the MAIC algorithm. To address this issue, we first applied an extreme gradient boosting (XGBoost) regression model to impute missing AOD values using a comprehensive set of predictors, including cloud fraction, fire counts, normalized difference vegetation index, land use and land cover, digital elevation, road proximity, socio-demographic indicators, and meteorological variables from ERA5 reanalysis (such as temperature, relative humidity, wind speed, and boundary layer height). All predictor variables were spatially regridded to a $1\text{-km} \times 1\text{-km}$ resolution using nearest-neighbor interpolation. In the second stage, the gap-filled AOD data, together with meteorological and land-use predictors, were used to estimate daily surface-level ambient $\text{PM}_{2.5}$ concentration using a random forest (RF) regression model at the same spatial resolution. The model was trained and validated using ground-based continuous monitoring data from the Central Pollution Control Board (CPCB) and demonstrated high predictive performance even at daily scale ($R^2 = 0.86$; $\text{RMSE} = 23.46 \mu\text{g}/\text{m}^3$), ensuring spatially consistent and reliable ambient $\text{PM}_{2.5}$ exposure estimates across India. Detailed descriptions of the

algorithmic framework and model performance were reported in our earlier work (Katoch et al., 2023). The resulting data set provided a bias-corrected, spatially continuous ambient $\text{PM}_{2.5}$ concentration (in $\mu\text{g}/\text{m}^3$) over India, significantly improving exposure representation in regions with persistent cloud cover or missing AOD values; especially after including the in situ observations for training and validation purposes. Satellite-retrieved data was integrated into the ML models to ensure its capacity to capture the complicated spatiotemporal variations of BC over India. Observations with high cloud cover, extreme solar zenith angles, and other quality flags that may enhance uncertainties were filtered out to improve data accuracy (www.earthdata.nasa.gov/learn/find-data/near-real-time/omi).

2.3. Meteorological Variables

The meteorological parameters used in the study were derived from ERA-5 reanalysis data (cds.climate.copernicus.eu/data-sets/reanalysis-era5-single-levels?tab=overview) produced by the European Center for Medium-range Weather Forecasts (ECMWF) (www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5). We obtained hourly databases for the key meteorological variables that are reported to affect aerosol dynamics, namely relative humidity, temperature, planetary boundary layer height, precipitation, and the u- and v-components of wind. For temporal alignment with the CTM-derived model output and the satellite-derived aerosol databases, which were obtained at a daily scale, we aggregated these hourly databases into daily data and fit them into the model. Moreover, the spatial resolution (at a quarter-degree scale) of ERA-5 databases was consistent with other predictor variables.

2.4. Demographic Data

The database for population density was obtained from the WorldPop website (hub.worldpop.org/project/categories?id=18) for the Indian region during the study period (2015–21). The WorldPop data set provides population density information at a high spatial resolution of 1-km $\times$ 1-km on an annual basis. This database was resampled to a quarter-degree resolution using a spatial aggregation method, where the population density values within each quarter-degree grid cell were averaged to derive a representative population density for that grid.

2.5. Machine Learning (ML) Models

We employed six ML models namely Extra Tree Regressor (ETR), Random Forest (RF), Decision Tree Regressor (DT), Light Gradient Boosting Machine (lightbm), Gradient Boosting Regressor (GBR), and Least Angle Regressor (LAR) to predict daily f_{BC} at $0.25^\circ \times 0.25^\circ$ spatial scale for the study period using a range of predictor variables. The six ML models used for the assessment were ensemble learning techniques for classification. The methods allow processing a sizable number of input features (variables) and ranking them according to their relevance (Zhou et al., 2022). Each data set is matched based on spatial grid cells (centroid) and dates to form a unified spatio-temporal dataframe containing all predictors and the target variable (BC mass fraction). Figure S4 in Supporting Information S1 summarized the data processing, model construction, and algorithm development in detail.

2.6. ML Model Training and Testing

The daily data for the base year 2015 were randomly divided into two parts. 70% of the data sets (221,472 data points) were used for training the model, while the remaining 30% (94,917 data points) were set aside for testing the model. We adopted this approach due to the smaller data set size, allowing for a representative sample to evaluate model performance. To train the model, we selected a series of predictor variables, namely population density, temperature, horizontal and vertical wind, relative humidity (RH), boundary layer height, AOD, precipitation, $\text{PM}_{2.5}$, NO_2 , and O_3 (Table S1a in Supporting Information S1). These variables were chosen for model training because meteorology, demography, other criteria pollutants, and primary $\text{PM}_{2.5}$ strongly influence the BC mass concentration, which was simulated using the CTM framework. We tested predicted daily scale f_{BC} from each of the six ML models against the remaining 30% of the data sets from CTM-derived f_{BC} for the base year 2015 across India (Venkataraman et al., 2019). To enhance the robustness of the model performance and ensure the generalizability in its prediction efficiency, we employed a 10-fold cross-validation (CV) technique, where the training data was divided into 10 subsets, training the model on 9 subsets and validating it on the 10th. This process was repeated 10 times with different random splits, and the aggregated predictions were then used to

evaluate model performance. The normalized feature importance plot (Figure S3 in Supporting Information S1) ranked the weighted predictive efficiency of each variable contributing to the model and contextualized model optimization. Additionally, the heatmap (Figure S5 in Supporting Information S1) illustrated the correlation among the variables and indicated no multi-collinearity among the predictors.

Among the six ML models, the performance of each model was rigorously examined using the correlation between the test and measured data sets for the base year 2015. For this purpose, key performance metrics, including R^2 , root mean squared error (RMSE), and mean absolute error (MAE), were used to gauge each model's accuracy and predictive power. The ETR emerged as the top performer compared to the others, exhibiting the highest R^2 value and the lowest RMSE and MAE among the evaluated models (Table S1c in Supporting Information S1). This algorithm's superior performance was further confirmed through a 10-fold CV, which provided a robust estimate of its generalizability and minimized the risk of overfitting. Additionally, residual analysis was conducted to ensure that the model's predictions remained unbiased to detect any potential anomalies or patterns in the prediction errors (Berrouachedi et al., 2019). The ETR's ability to handle complex variable interactions and its overall robustness to different data subsets solidified its selection as the optimal model for predicting f_{BC} (Schmid et al., 2023). Text S1 and S2 in Supporting Information S1 described the ETR model, its functionalities, and the methodology of hyperparametric tuning on the model algorithm.

2.7. Estimation of BC Mass Concentration

The ETR-model algorithm was used to predict f_{BC} for the subsequent years (2016–21, Figure S6 in Supporting Information S1), with the input predictor variables remaining consistent and utilized for model development in 2015. The model's generalization was assessed by examining its prediction accuracy over this extended period, ensuring it maintained consistent performance across multiple years without the need for re-training. This approach leveraged the model's ability to generalize the relationship between the predictor variables and f_{BC} established during the 2015 training phase, allowing for robust trend analysis and facilitating an understanding of long-term changes in f_{BC} over time. The consistency in model setup and prediction strategy ensured that any observed trends could be attributed to actual changes in the predictors rather than variations in model configuration. Later, the predicted f_{BC} was integrated with our recently developed ambient $PM_{2.5}$ concentration at a 1-km $\times$ 1-km spatial resolution (Katoch et al., 2023) to obtain the BC mass concentration at a higher spatial resolution for the entire study period (2015–2021). We assumed similar BC mass fractions within each quarter-degree grid cell. We overlaid the gridded population databases (2015–2021) onto the BC mass concentrations and estimated the population-weighted mean BC exposures across the urban and rural regions of the states, which were then used in the health burden and exposure disparity assessments.

We acknowledged that the CTM simulations carry inherent uncertainties in predicting ambient $PM_{2.5}$ concentration and its chemical constituents, particularly for BC, and that CTM outputs often underestimate exposure levels relative to in situ observations (Lyu et al., 2024). To address these limitations, we incorporated CTM-derived BC estimates together with other relevant predictors into the ML models to estimate f_{BC} , which was directly validated against CTM-derived f_{BC} for the base year 2015. In addition, we independently validated the ML-derived BC mass concentrations using ground-based BC measurements across India for the contemporary study period (Table S2 in Supporting Information S1). These hybrid approaches to validate our model-derived BC estimates enabled us to retain the large-scale spatial consistency of CTM outputs while improving local accuracy and spatial resolution through ML-based modeling. To evaluate the performance and generalizability of the ETR model, we conducted a multi-year validation (2015–2021) using coincident observations from 36 ground-based BC measurements compiled from the published literature from India (Table S2 in Supporting Information S1). As shown in Figure S7 of Supporting Information S1, these sites were distributed across the country and represented diverse geographic, climatological, and socio-demographic settings spanning both urban and rural regions. Together, these additions demonstrate that the validation data set captures both urban and rural environments, thereby reducing the likelihood of systematic bias toward a particular settlement type. For each observation site, the centroid of the location was matched to the nearest 1-km grid cell of the ETR modeling framework, and the corresponding mean BC concentration over the study period was extracted to compute the validation statistics. The strong agreement between modeled and observed BC concentrations ($R^2 = 0.82$; RMSE = $1.77 \mu\text{g}/\text{m}^3$; see Figure 1b in Section 3) indicated that the ML-derived BC estimates effectively captured the spatio-temporal variability of BC across India over the full study period (2015–2021), supporting their application in subsequent exposure disparity and health burden assessments.

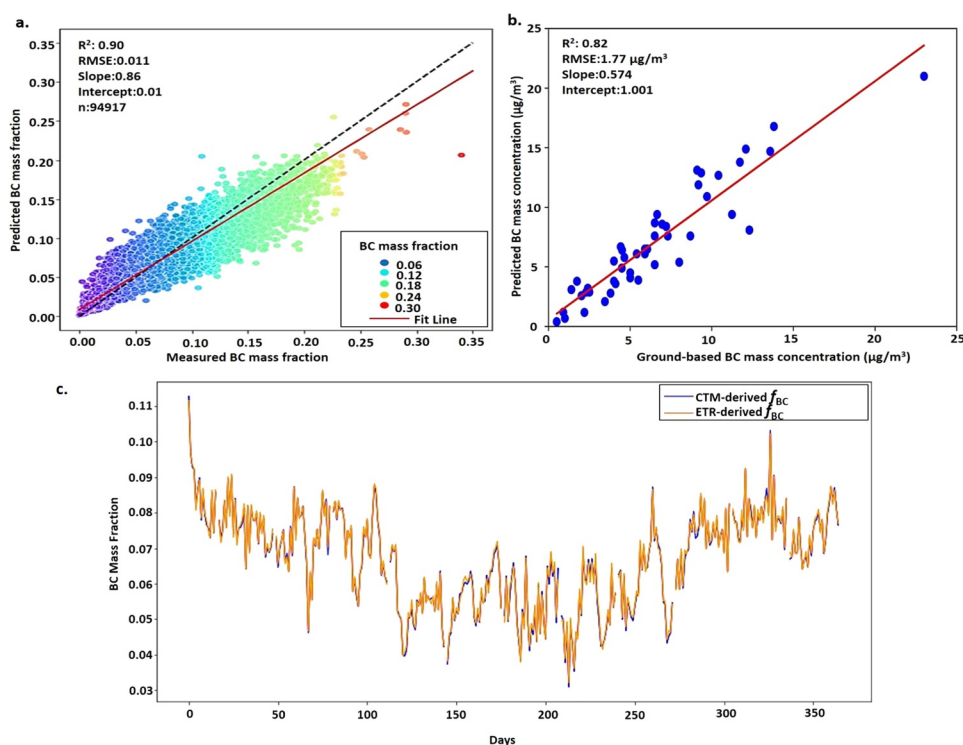


Figure 1. (a) Correlation between ETR-model predicted f_{BC} with the CTM-derived f_{BC} for the base year 2015 ($n = 94,917$). The black dotted line represents the best fit ($R^2 = 1$) between the predicted and measured BC mass fractions, and the red solid line denotes the model-fitted line. Panel (b) correlation between ETR-model predicted f_{BC} with the ground-based BC measurements from 36 different regions across India. The red dotted line represents the model-fitted line between the predicted and measured f_{BC} . Panel (c) daily time series of ETR-model predicted (orange line) and CTM-derived (blue line) f_{BC} in 2015.

2.8. Estimation of Health Burden Attributable to BC Exposure

Recently published meta-analyses provided the pooled effect estimates (β -coefficients) for all-cause, cardiovascular disease (CVD), and respiratory mortalities per unit increase in BC exposure (Zhu et al., 2023); which were used to first estimate relative risks (RRs) for these three types of mortalities attributable to population-weighted BC exposure across the states (see Equation 1). These cause-specific RR estimates were integrated with epidemiologic and demographic attributes to estimate BC exposure attributable mortality burdens across the states from 2015 to 2021. We obtained the adult population (>25 years) estimates for the states between 2015 and 2021 from the census projected report, which forecasts the age-distributed population from 2011 to 2036 at 5-year intervals (Peoples' Archive of Rural India (PARI) report, n.d.). For the intermediate years, we interpolated the statistics using bilinear interpolation method. On the other hand, we obtained the all-cause, CVD, and respiratory mortality rates (per 100,000 population; with 95% Uncertainty Intervals, UIs) for the states, for adult population, both gender combined, over the study period from the GBD-India portal (Murray, 2024).

In the meta-analyses, the β coefficients were adjusted for potential confounding by relevant socioeconomic and demographic covariates, as well as by ambient $PM_{2.5}$ and other co-pollutants. Consequently, these resulting cause-specific risk estimates could be solely attributed to per-unit changes in BC exposure. These β coefficients quantify the incremental mortality risk associated with elevated BC concentrations, and we used them within an empirical framework to estimate RRs for all-cause, CVD, and respiratory mortalities across the states and years,

$$RR = e^{\beta(x-x_{ref})} \quad (1)$$

where β represents the effect estimates or increase in mortality risks attributable to a unit increase in BC exposure, reported as 0.026 (95% Confidence Intervals, CIs: 0.008–0.044) for all-cause mortality, 0.009 (0.007–0.011) for CVD mortality, and 0.028 (0.007–0.048) for respiratory mortality, respectively; x is the population-weighted

mean BC exposure across the states and years; and x_{cf} is the counterfactual exposure level for BC. Note that counterfactual exposure levels are ideally grounded in established guidelines or theoretical minimum risk exposure levels. While such counterfactuals are well defined for total ambient $PM_{2.5}$ in the GBD framework (Murray, 2024), no theoretical minimum risk exposure level has yet been established for black carbon or other chemical constituents of ambient $PM_{2.5}$ in the existing epidemiological literature. In the absence of an evidence-based or guideline-recommended counterfactual for BC, we adopted a data-driven and conservative approach by using the lowest observed annual population-weighted BC exposure within India for each study year (2015–2021) as the counterfactual level. This approach is consistent with prior air-pollution health burden assessments for pollutants lacking defined theoretical thresholds and avoids extrapolation beyond the observed exposure range (Brauer et al., 2022). Importantly, using an internally derived annual minimum also accounts for inter-annual variability in background BC levels and provides a stable reference for temporal and spatial comparisons. We acknowledge that this choice may influence the absolute magnitude of the estimated attributable burden, but it does not affect the relative comparisons across population subgroups or the disparity patterns that are central to this study. Further, we estimated the population attributable fraction (PAF) of each grid cell,

$$PAF = 1 - \frac{1}{RR} \quad (2)$$

The number of premature deaths (*Mort*) per year attributable to BC exposure across the states was calculated as $Mort = PAF \times Pop \times MR$, where *Pop* is the total *at-risk* adult population, and *MR* is the all-cause, CVD, and respiratory mortality rates, respectively. In the meta-analyses, the 95% CIs of the cause-specific β estimates were derived from 1,000 model runs for each quantity of interest, with the means have been taken as the point estimates and the 2.5th and 97.5th percentiles of the resulting distributions defining the 95% CIs (Zhu et al., 2023). Using the analytical framework of Equations 1 And 2, the lower and upper CIs were used to estimate the 95% UIs for the cause-specific RRs and its associated PAF estimates. For state-level BC-attributable mortality burden estimates, we computed corresponding 95% UIs by propagating uncertainty from the lower and upper bounds of the cause-specific β coefficients and *MR*, which were multiplied to obtain the lower and upper UI-bounds, respectively. Note that, uncertainty attributable to demographic estimates could not be incorporated, as state-level UIs for these determinants were not available for the study period; consequently, these sources of uncertainty were not included in the overall UI-assessment. This comprehensive analysis provided insights into the subnational variation of BC-attributable health impacts and identifying regions with higher exposure-related mortality burdens. We reported the mortality (per 100,000 population) estimates in the main paper and the absolute burden in the Supporting Information S1.

We acknowledge that differences in exposure definition, measurement approach, spatial scale, and source composition may affect the transferability of β coefficients, derived from epidemiological meta-analyses with model-derived BC exposure estimates. However, the meta-analysis by Zhu et al. (2023) harmonized effect estimates across diverse BC metrics (optical BC, black smoke, elemental carbon) and providing a harmonized exposure-response relationship suitable for population-level health assessments. Moreover, many underlying epidemiological studies themselves relied on modeled or proxy-based BC exposures when monitoring data were limited, reflecting standard practice in large-scale health burden assessments. Secondly, our high-resolution BC exposure is consistent with the outdoor exposure metrics used in the underlying epidemiological studies. While some individual-level exposure misclassification is unavoidable, such error is expected to produce conservative risk estimates in population-level analyses, which has been taken care of in the 95% UI estimates. Lastly, following the GBD framework, we assumed transferability of exposure-response functions in the absence of large India-specific cohorts. While this standard approach enables policy-relevant first-order estimates, some uncertainty may remain due to regional differences in BC composition, sources, and exposure assessment, warranting cautious interpretation and future India-specific cohort validation. Future cohort-based studies, particularly in high-BC exposure regions with socio-demographic contexts comparable to India, are essential to reduce existing uncertainties and strengthen the evidence base for region-specific concentration-response relationships. We classified the states into three socio-demographic indices namely low SDI (≤ 0.53), middle SDI (0.54–0.6), and high SDI (> 0.6); consistent with the classification used in the GBD-India study (Balakrishnan et al., 2019) using a combination of log-distributed per capita income, mean education (15 years or above), and fertility rate in women (< 25 years). We also estimated total $PM_{2.5}$ attributable all-cause mortality over the study period, using the RR estimates from another meta-analysis. We performed sensitivity analyses to estimate the avoidable deaths in each

state for a reduction in BC exposure by 25%, 50%, and 75%, respectively (Table S3 in Supporting Information S1).

2.9. Isolating the Effects of Determinants on the Estimated Health Burden

Since BC-attributable deaths depend on changes in its exposure, age-distributed population, and cause-specific MRs across the years, their individual effects were segregated to understand the relative importance of each socio-demographic determinant on health burden trends between 2016 and 2021 (Table S4 in Supporting Information S1). To achieve this, three additional sets of assessments were conducted. First, we estimated BC-attributable deaths for 2016 to 2021 by changing only the MRs while keeping demographics and BC exposure constant at their 2015 levels. Secondly, we altered only the BC exposure while keeping the other factors constant as per their 2015 levels. Lastly, we changed only the demographics, maintaining the other two factors unchanged. The differences in BC exposure attributable burden estimates (relative to base year 2015) in these three additional sets of estimates reflect the individual effect of that particular determinant, whose actual changes from 2016 to 2021 were considered (Chowdhury et al., 2020).

2.10. Disparity in BC Exposure Among the Sub-Populations of India

We analyzed the socio-demographic information from the National Family Health Survey (NFHS, n.d.), which had a two-stage sampling design: first, districts were selected, and then the clusters (villages in rural areas and census enumeration blocks (CEBs) in urban areas) were chosen from each district. NHFS-4 covered a total of 28,527 clusters from 640 districts and interviewed 811,808 respondents, while NHFS-5 covered 30,197 clusters from 707 districts of India and a total of 825,954 individuals (some larger districts were divided to create smaller districts during this period, increasing the total number of districts). A detailed description of the NFHS survey framework is documented in Text S3 of Supporting Information S1.

Demographic data from the NFHS include individuals' gender (male and female), wealth-index (five subgroups namely richest, richer, middle class, poorer, and poorest), primary cooking fuel-types [*“clean”* for households using electricity, Liquefied Petroleum Gas (LPG), and natural gas as primary fuel; and *“solid”* for households using biomass, kerosene, coal or lignite, charcoal, wood, straw/shrubs/grass, agricultural crop, and animal dung], and caste [General, Other Backward Class (OBC), and Scheduled Caste and Scheduled Tribes were combined as (SC + ST)], which is equivalent to “Ethnicity” in the Western world.

We first extracted the annual mean BC exposure for households based on their geo-locations (centroid) within 2 and 5 km buffers for urban and rural clusters of the latest NFHS rounds [NFHS-4 (2015–16) and NFHS-5 (2019–21)]. Nationally, each subgroup's population-weighted exposure (PSE) was estimated by a weighted average of cluster-level BC concentrations within a given region, with the assigned *“sample weight”* being the proportion of sub-population (representative of actual census-demography) within each cluster estimated by survey data (DHS-India). If there are m clusters in a region, PSE is defined as,

$$PSE_{ik} = \frac{\sum_{j=1}^m C_{ij} \times P_{ijk}}{\sum_{j=1}^m P_{ijk}} \quad (3)$$

where PSE_{ik} is the PSE to black carbon for the k th population subgroup of the i th region, C_{ij} is the BC concentration at the j th cluster of the i th region, and P_{ijk} is the estimated proportion of the k th population subgroup at the j th cluster of the i th region. We accounted for the sample weight of every cluster reported by NFHS to estimate PSE. We excluded the null values from our analysis. We estimated absolute disparity *“within each subgroup”* across the urban and rural regions of India by considering the difference between the 95th and 5th percentile exposure levels for both NFHS rounds (Table S5a in Supporting Information S1). Additionally, we estimated the *“across-subgroup”* relative disparity (denoted in-%) in the urban and rural regions of the NFHS rounds by calculating the differences in population-weighted mean exposures across the sub-populations, divided by the PSE among overall population (Table S5b in Supporting Information S1). For analytical purpose and better interpretability, we took the difference in PSEs among richer, middle class, poorer, and poorest subgroups with respect to the PSE estimate among the richest subgroups. Similarly, the PSEs of OBC and SC + ST subgroups were compared with the PSE of General-caste subgroup in urban and rural regions. We estimated the changes in absolute and relative disparities between the NFHS time points (2015–16 to 2019–21). We acknowledge that the

changes in BC exposure disparities over the time points were the cumulative results of changes in both demographic and BC exposure distributions from NFHS-4 to NFHS-5 across the sub-populations. Therefore, we conducted two sensitivity analyses to disentangle the respective contributions of demographic structure and BC exposure distribution on the changes in absolute and relative disparities across demographic subgroups. In the first analysis, we only changed the demographic distribution from NFHS-4 to NFHS-5 while holding BC exposure constant at its NFHS-4 distribution. In the second analysis, we updated the BC exposure distribution from NFHS-4 to NFHS-5 while keeping the demographic distribution fixed at NFHS-4 levels. The net change (positive or negative; Tables S5c and S5d in Supporting Information S1) in subgroup-specific disparities with respect to the actual disparity estimate during NFHS-4 reflected the relative contributions of each determinant across NFHS survey periods, stratified by urban and rural regions.

We then performed the Kolmogorov-Smirnov (KS)-test (Massey, 1951) to assess the statistical significance (determined by $p < 0.001$) across the exposure distributions of the subgroups (Table S6 in Supporting Information S1). In our case, it involved BC exposure over India. This test was based on the maximum difference between the cumulative distribution functions (CDFs) of the two distributions being compared. The KS-statistics measures this difference and was used to estimate whether the two distributions were significantly different from each other. The KS test statistic was calculated by $D_n = \max|F_{n1}(x) - F_{n2}(x)|$; where F_{n1} and F_{n2} are the empirical CDFs of two samples, x represents each point in the distribution, and the test statistic D_n is a measure of the maximum difference between the two CDFs. In our assessment, we estimated the difference between BC exposures from 2016 to 2021 while keeping 2015 distribution as the reference. Larger estimates of the KS-statistics denote higher statistical differences between the two BC exposure distributions, and vice-versa.

2.11. Estimation of BC Attributable Mortality Across the Population Subgroups

As described in Section 2.8, the RR estimates for all-cause, CVD, and respiratory mortality were expressed as a function of long-term BC exposure and its deviation from the counterfactual level, scaled by the meta-analysis-derived β coefficients representing the risk increments per unit increase in BC exposure (Equation 1). These β coefficients were obtained from cohort-based epidemiological models adjusted for individual-level age, sex, and a range of relevant socioeconomic and demographic factors (Zhu et al., 2023); therefore, the resulting cause-specific unit risk increments could be considered representative and same across diverse demographic subgroups. However, the RR estimates for these mortality-types may vary across sub-populations due to the differences in their PSEs across urban and rural regions of the states. Based on these analytical considerations, we first estimated subgroup-specific RRs for all-cause, CVD, and respiratory mortalities attributable to long-term BC exposure. We subsequently computed the corresponding attributable fractions (AFs) for each mortality-cause,

$$RR = e^{(\Delta PM_{2.5}) \times \beta}; AF = \frac{RR - 1}{RR} \quad (4)$$

Here $\Delta PM_{2.5}$ is the difference between population-weighted annual BC exposure of a given population subgroup in the urban and rural regions and the lowest estimated BC level (x_{cf}) for all populations, considered as the counterfactual exposure level; β is the meta-analyses-derived unit-risk increments for all-cause, CVD, and respiratory mortalities, respectively, considered same for all sub-populations. The demographic information from the NFHS was used to derive the fractions of *at-risk-population* across the demographic subgroups, incorporating the cluster-specific sample weights, while the same all-cause, CVD, and respiratory mortality rates (MRs) were used for all sub-populations. The GBD framework was used to estimate mortality burden across each population subgroup.

3. Results

3.1. BC Exposure in India

We first assessed the efficiency of six ML models in predicting f_{BC} (Figure S8, Table S1a in Supporting Information S1), among which the ETR showed the strongest agreement ($R^2 = 0.92$) and the least bias (root mean square error of 0.011 and mean absolute error of 0.007) between the ML-predicted and CTM-derived f_{BC} for the base year 2015 (Figure 1a). The ETR model-predicted f_{BC} daily time series also showed a robust match with CTM-derived f_{BC} (Figure 1c). We then predicted f_{BC} using only the ETR model for the period 2016–2021 (Figure

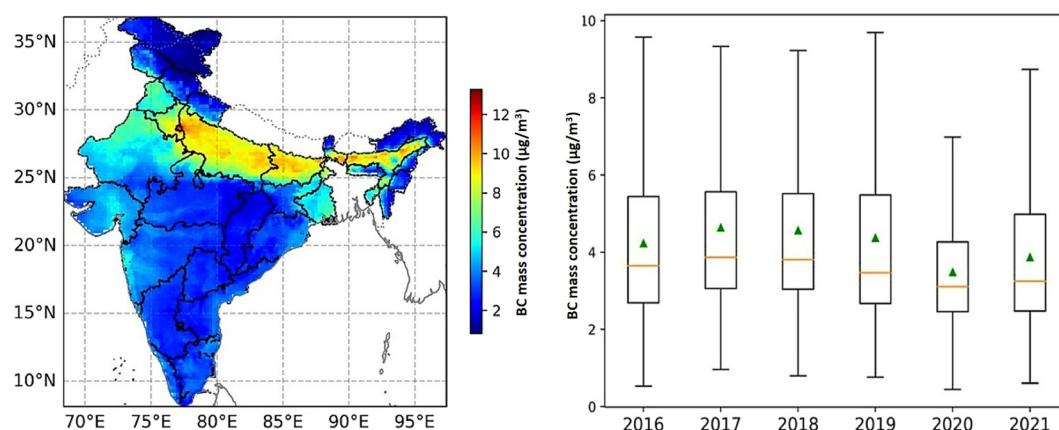


Figure 2. Spatial variation (left) of black carbon (BC) mass concentration across India from 2016 to 2021. The estimated BC mass concentration for each year between 2016 and 2021 is documented in Figure S2 of Supporting Information S1. Variation (5th to 95th percentile) of BC mass concentration between 2016 and 2021 across India (right). The green triangle denotes the mean BC concentration, and the orange horizontal line denotes the median value of the distribution, while the outliers are excluded.

S6 in Supporting Information S1) and converted f_{BC} to BC mass concentrations for the same period (Figure S2 in Supporting Information S1). Our independent evaluation (based on reported data compiled in Table S2 of Supporting Information S1) against 36 in situ BC concentrations revealed a statistically significant correlation ($R^2 = 0.82$) and an lower RMSE of $1.77 \mu\text{g}/\text{m}^3$ (Figure 1b). Although we found our independent validation to be robust for exposure assessment, we note that regional or seasonal validation is not possible due to the lack of adequate in situ observations at the desired scales.

The spatial patterns of BC concentrations averaged over the period 2016–21 (Figure 2) revealed a strong north–south decreasing gradient. BC exposure exceeded $8 \mu\text{g}/\text{m}^3$ in the IGP states, Assam, and the Mumbai industrial region (on the west coast of India). Within the IGP, we found the highest exposure ($>12 \mu\text{g}/\text{m}^3$) in the national capital, Delhi. Exposure was usually lower than $4 \mu\text{g}/\text{m}^3$ in other regions of India (Figure S2 in Supporting Information S1). BC exposure peaked in 2018 (the all-India population-weighted average was $4.6 \pm 2.4 \mu\text{g}/\text{m}^3$), while it dropped significantly (KS-statistics: 0.225, $p < 0.001$; Table S6 in Supporting Information S1) in 2020 ($3.5 \pm 1.7 \mu\text{g}/\text{m}^3$), particularly across the central and eastern states, reflecting the impact of sudden emission reductions for several months during the COVID-19-induced lockdown in India. We estimated a large urban-rural gap in BC exposure across the years (Figure S9 in Supporting Information S1), especially at higher BC levels ($>7.5 \mu\text{g}/\text{m}^3$; KS-statistics: 0.061–0.463, $p < 0.001$).

3.2. BC-Attributable Premature Mortality Burden in India

We estimated BC-attributable all-cause, CVD, and respiratory mortalities of 133 (95% Uncertainty Intervals, UIs: 113–155), 7 (5–9), and 9 (5–14) per 100,000 population, respectively. In absolute terms, annual BC-attributable premature deaths were 1,207,446 (965,957–1,448,935), 55,176 (46,900–63,452), and 65,951 (54,080–77,822), respectively, in India during 2016–21. These estimates indicated that these many premature deaths could be averted annually across India under a scenario of complete elimination of BC exposure across the states. We then isolated the effects of individual drivers namely BC exposure, exposed population, and the all-cause, CVD, and respiratory mortality rates (MRs) on the changes in aggregated BC-attributable deaths in India. If the demography and MRs remained unchanged throughout 2016 to 2021, BC-attributable all-cause, CVD, and respiratory mortalities would have been lower by 26.1%, 18.5%, and 17.5%, respectively, due to reduction in BC exposure in 2021 as compared to their actual estimates (Table S7 in Supporting Information S1). If the MRs and BC exposure remained unchanged, then the mortalities would have been higher by 17.2%, 16.9%, and 22.3%, respectively, attributable to the changes in demographic structure in 2021 relative to the actual estimates. And lastly, for unchanged demography and BC exposure, the all-cause, CVD, and respiratory mortalities would have been higher by 6.5%, 9.6%, and 8%, respectively, in 2021 due to changes in MRs only across the states. This analysis indicated that population growth was the primary driver of the increasing health burden across India, followed by

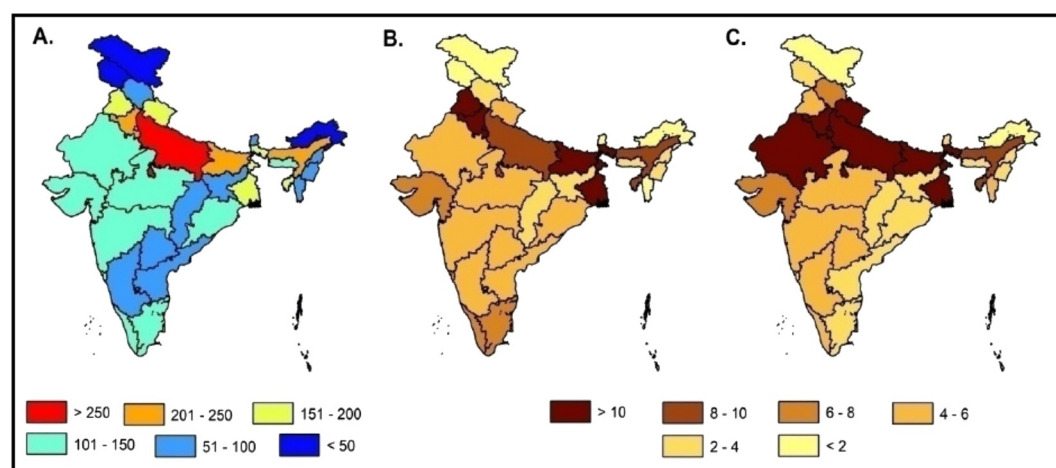


Figure 3. BC-attributable (a) all-cause, (b) cardiovascular disease (CVD), and (c) respiratory mortalities (per 100,000 population) across the states, averaged over 2016 to 2021. Figure S5 in Supporting Information S1 lists the BC-attributable mortality burden for each year.

risers in cause-specific MRs, while reductions in BC exposure mitigated mortality burdens and partially offset the increases attributable to the other socio-demographic determinants over the years.

The states with a lower socio-demographic index (SDI) bore the largest share of the mortality burden (54%, 43%, and 62% for all-cause, CVD, and respiratory mortalities, respectively), followed by middle SDI states (25%, 31%, and 23%), and high SDI states (21%, 26%, and 15%) (Figure S10 in Supporting Information S1). Uttar Pradesh (293 deaths per 100,000 population), Delhi (235 per 100,000), and Bihar (230 per 100,000) exhibited the highest BC-attributable all-cause mortality burden (Figure 3a), while the central and western regions experienced moderate mortality burdens (100–180 deaths per 100,000). The IGP states and Assam suffered from the highest BC-related CVD and respiratory mortalities (>10 deaths per 100,000), while a strong north–south decreasing gradient was evident across India. The observed temporal variations in mortality burdens over the years aligned with the changes in BC exposure across the states (Figure S10 in Supporting Information S1). For instance, BC-attributable CVD and respiratory mortalities showed increasing trends between 2016 and 2018, followed by a reduction from 2019 to 2020; however, they significantly increased again in 2021 across the states. Moreover, we estimated a higher ratio of BC-attributable CVD to all-cause mortality in the southern peninsular region compared to other parts of India, and a higher ratio of BC-attributable respiratory mortality to all-cause mortality across the IGP and other northern states (Figure S11 in Supporting Information S1).

3.3. Disparity in BC Exposure and Attributable Mortality Across Demographic Subgroups

We first estimated the “*within-subgroup*” absolute disparity in urban and rural regions across the NFHS rounds and noted several key features. First, we observed a larger disparity among the urban population (5.9–8.7 and 6.6–10.5 $\mu\text{g}/\text{m}^3$ in NHFS-5 and NHFS-4, respectively) than in the rural population (6–6.8 and 6.3–9.1 $\mu\text{g}/\text{m}^3$ in NHFS-5 and NHFS-4, respectively) across most demographic subgroups (Figure 4), due to the more heterogeneous distribution of sources in urban areas. Secondly, in urban areas, the disparity increased with socioeconomic status (implying a larger disparity for the rich than for the poor), while it showed a reverse trend in rural areas. Third, we estimated higher absolute disparity among individuals belonging to the General and OBC-castes than among SC + ST subgroup. Lastly, we found a larger reduction in absolute disparity from NHFS-4 to NHFS-5 in the rural regions, among females, wealthier subgroups, and General and OBC-caste subgroups as compared to their demographic counterparts. Conversely, the disparity declined by a larger margin among deprived subgroups and SC + ST in urban areas.

We then estimated the relative disparity in urban and rural regions (Table S5b in Supporting Information S1) to identify which subgroups were exposed to higher BC concentrations than their demographic counterparts. During NHFS-5, BC exposure was higher among the poorest and poorer subgroups as compared to the rich and middle class subgroups. Among the caste-based subgroups, the General and OBC-castes faced higher BC levels than

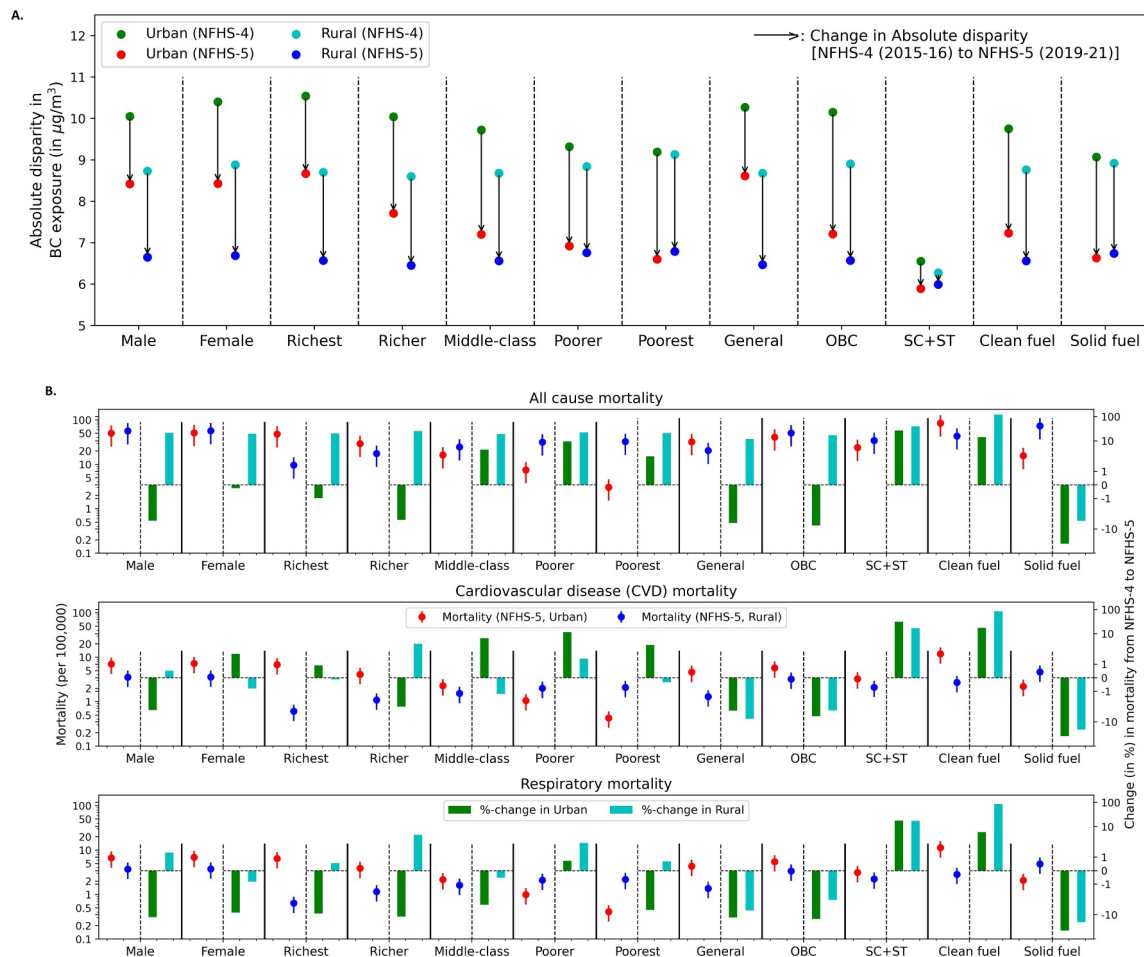


Figure 4. (a) Absolute disparity (the difference between 95th and 5th percentile) in BC exposure within each subgroup in the urban and rural regions of NFHS rounds. The direction of vertical arrows indicates the change in absolute disparity (in $\mu\text{g}/\text{m}^3$) from NFHS-4 (2015–16) to NFHS-5 (2019–21). We perform the Kolmogorov-Smirnov (KS)-test to estimate whether the exposure frequency distributions across the population subgroups are statistically different (value of KS-statistics, determined by $p < 0.001$). The mean exposures (± 1) of the subgroups in the urban and rural regions are documented in Table S5b of Supporting Information S1. Panel (b) estimated BC attributable all-cause, CVD, and respiratory mortalities in NFHS-5 (red and blue circles) and their changes (in-%) from NFHS-4 (green and cyan bars) across the population subgroups stratified by gender, wealth, caste, and cooking-fuel types in the households for the urban and rural regions. The circles denote the mean estimates of BC-attributable mortality burden and the whiskers are the associated 95% uncertainty intervals. The Y-axes are in logarithmic scale.

SC + ST subgroups. Solid-fuel users experienced greater BC exposure than clean-fuel users in both urban and rural areas. During NFHS-4, the deprived subgroups, General and OBCs suffered from higher BC exposure as compared to their demographic counterparts. Regarding gender, the differences in exposure levels for males and females were not significant in both NFHS rounds. Across the subgroups, the relative disparity also decreased from NFHS-4 to NFHS-5. Our sensitivity assessment dictated that reductions in BC exposure among demographic subgroups over NFHS time points have reduced both absolute and relative disparities (Tables S5c and S5d in Supporting Information S1), while changes in demographic distributions have played modest roles.

We finally analyzed the disparity in BC-attributable mortality burdens across the subgroups. We found contrasting trends in the disparity of BC-attributable all-cause, CVD, and respiratory mortalities across wealth and caste-based subgroups in urban and rural regions during NFHS-5. For example, the BC-attributable all-cause, CVD, and respiratory mortalities were higher among richest in urban areas (48, 7, and 6 deaths per 100,000 population, respectively) than among the poorest (3, 1, and 1 death per 100,000 population, respectively). We observed an opposite trend in rural regions (Figure 4b). Similarly, clean-fuel users suffered from larger BC-attributable mortalities (85 vs. 16, 12 vs. 2, and 11 vs. 2 deaths per 100,000 population, respectively, for all-cause, CVD, and respiratory mortality) in urban regions as compared to solid-fuel users. In contrast, solid-fuel

users experienced a higher mortality burden in rural areas. Across caste-based subgroups, the mortality burden was higher for OBCs, followed by SC + ST, and the General-caste sub-population.

Overall, BC-attributable all-cause mortality increased by 12%–121% in rural areas across most of the subgroups between NHFS-4 and NHFS-5, except among solid-fuel users, where it decreased by 4.7%. Conversely, BC-attributable all-cause mortality declined among males, wealthier subgroups, and solid-fuel users in urban regions by 4.3%–41%; while it increased among their demographic counterparts by 2.3%–27% between NHFS-4 and NHFS-5. Similar patterns of temporal changes in BC-attributable CVD and respiratory mortalities across the caste and cooling-fuel type categories were observed. For instance, the mortality burden increased among SC + ST subgroup by 16%–31% and among solid-fuel users by 5%–85%, respectively, while it decreased among their demographic counterparts (Figure 4b). However, we observed mixed patterns in temporal changes in BC-attributable CVD and respiratory mortalities among the gender and wealth-based subgroups.

4. Discussion and Conclusions

In this study, we present a comprehensive assessment of long-term BC exposure, its spatio-temporal variability at high resolution, and its associated health impacts across major demographic subgroups in India. We developed a novel, high-resolution BC data set using advanced ML techniques and integrated it with socio-demographic information from two nationally representative demographic surveys. We acknowledge that CTM frameworks inherently carry uncertainties in simulating ambient $PM_{2.5}$ and its chemical constituents including BC, and often underestimate concentrations relative to ground-based observations (Lyu et al., 2024). To address these limitations, we employed an ML-based hybrid framework in which CTM-simulated outputs were used as predictors alongside relevant meteorological, geospatial, and demographic variables to estimate f_{BC} . These estimates were subsequently combined with a satellite-derived ambient $PM_{2.5}$ data set (Katoch et al., 2023), calibrated against in situ observations, to generate BC concentrations across India. The strong calibration of the $PM_{2.5}$ data set with ground-based measurements, together with the use of relevant predictor variables (Table S1a in Supporting Information S1) and the best-performing ML algorithm, supports the robustness of the developed BC data set. Furthermore, we validated the modeled BC concentrations against 36 coincident ground-based measurements distributed across diverse geographic, climatological, and socio-demographic settings in both urban and rural regions of India. The strong agreement between modeled and observed BC concentrations (Figure 1b) indicated that the ML-derived estimates effectively captured the spatiotemporal variability of BC across India during the study period (2015–2021), supporting their application in subsequent health burden and exposure disparity assessments.

Note that a strong correlation between CTM-derived BC concentration with satellite-derived ambient $PM_{2.5}$ estimate for the baseline year of 2015 ($r = 0.93$, $p < 0.05$) reflects their physical linkage rather than methodological redundancy; which is also evident by high BC-to- $PM_{2.5}$ deaths ratio (>0.7) across the northern states, especially over the IGP region, while this ratio showed strong north-to-south decreasing gradient (Figure S12 in Supporting Information S1). Importantly, BC exhibited moderate to higher correlations ($r = 0.261$ – 0.962 , $p < 0.05$; see Table S1b in Supporting Information S1) with other model-simulated PM-components (Sadavarte & Venkataraman, 2014), indicating that it retains meaningful source-specific information associated primarily with combustion emissions. Note that in environmental justice assessments, exposure disparity is typically evaluated by integrating pollutant distributions with demographic patterns and comparing population-weighted exposures across demographic subgroups. Accordingly, although our model-derived BC is strongly correlated with satellite-derived $PM_{2.5}$, variation in BC exposure was independently integrated with demographic distributions to quantify across-subgroup disparities. This approach allows BC and $PM_{2.5}$ disparities to be interpreted separately, consistent with prior US-based studies (Clark et al., 2017; Rosofsky et al., 2018). While our previous work reported inequality in ambient $PM_{2.5}$ exposure (Sarkar et al., 2026), the present study focuses on disparities in BC estimate, one of the key combustion-related PM-components, to identify inequality-hotspots and vulnerable sub-populations for targeted policy actions. Importantly, the meta-analytic risk coefficients were adjusted for $PM_{2.5}$ and other co-pollutants (Zhu et al., 2023), enabling attribution of cause-specific mortalities specifically to BC exposure across the states and years. Therefore, the estimated burden and disparities should be interpreted as reflecting the combustion-related fraction of fine particulate pollution, which remains policy-relevant. Nonetheless, we applied the trained model (during 2015) to predict f_{BC} for 2016–2021 using the same predictor set, thereby assuming that the functional relationship between predictors and f_{BC} learned in 2015 is sufficiently stable for subsequent years. However, note that this approach did not assume invariant emissions; rather, interannual

changes in emissions and activity patterns (including the COVID-19 period) were partially reflected through the changes in time-varying predictors (satellite-derived NO_2 , $\text{PM}_{2.5}$, AOD, and meteorology). More importantly, the moderate to lower correlations among the predictors (Figure S5 in Supporting Information S1) depict that these variables were treated independently in the ML-models to predict f_{BC} , and no multi-collinearity among these variables signify their time-varying independent impacts on the outcome over the years. Nonetheless, our variable importance assessment (Figure S3 in Supporting Information S1) showed grater contributions from demographic and meteorological parameters to predict BC mass fractions, as compared to the co-pollutants. Therefore, any significant change in these predictors (with larger variable importance) may influence inter-annual trends in BC exposure and its attributable mortality burden. To evaluate potential non-stationarity effects, we conducted an independent multi-year validation (2015–2021) against the 36 ground-based BC measurement data sets across India, which showed strong overall agreement ($R^2 = 0.82$; $\text{RMSE} = 1.77 \mu\text{g}/\text{m}^3$), supporting the model's generalizability and robustness in capturing potential changes in emissions, source composition, activity patterns (such that COVID-related emission reductions), and the spatio-temporal variability in BC exposure across the years.

A recent India-specific assessment reported approximately 400,000 CVD deaths attributable to wintertime BC exposure across the IGP region (Verma et al., 2022), which is substantially higher than our national-scale estimates. This discrepancy is primarily attributable to differences in exposure magnitude, baseline mortality inputs, and temporal coverage. Verma et al. (2022) reported very high mean BC concentrations ($>20 \mu\text{g}/\text{m}^3$) during winter, whereas the annual population-weighted BC exposure over IGP states in our analysis ranged between 6 and $11 \mu\text{g}/\text{m}^3$. Compounded by higher BC exposure levels and greater β -coefficient in their study (0.0214 vs. 0.009 per $\mu\text{g}/\text{m}^3$), these translated into markedly elevated relative risks (1.5–4) for BC-attributable CVD mortality, as compared with more moderate RR range (1–1.2) estimated in our study. In addition, Verma et al. (2022) adopted higher MR (272 deaths per 100,000 population) derived from survey-based estimates for 2010 (Prabhakaran et al., 2016), whereas our study used the more recent and widely adopted GBD-based CVD mortality rate (Murray, 2024), which is substantially lower (178 deaths). We note that the β -coefficient used by Verma et al. (2022) was derived from a California-based longitudinal study, where BC exposure levels and population characteristics differ substantially from the IGP context. Application of such coefficients may contribute to higher burden estimates. Conversely, our analysis employed a pooled β -estimate from a recent meta-analysis that synthesized evidence across diverse high-exposure settings, including urban Chinese populations with socio-economic and demographic conditions more comparable to India. We therefore consider the unit risk estimate used in this study to be more regionally representative, while acknowledging that some uncertainty in cross-population transferability remains. Importantly, their analysis was limited to wintertime BC simulations and did not account for seasonal variability, particularly the substantial decline in BC concentrations during the monsoon period. Because the GBD comparative risk assessment framework is designed for long-term (chronic) exposure, the use of season-specific exposure fields may introduce inherent uncertainty in annual attributable burden estimates. In contrast, our analysis integrates multi-year, annual-average BC exposure, thereby providing a more policy-relevant estimate of chronic population risk at the national scale.

Policy implications: Our study is novel and critical for policy in multiple aspects. First, our long-term systematic database at such a high spatial scale ($1\text{-km} \times 1\text{-km}$) enables an understanding of the local BC hotspots and tracking progress in mitigating anthropogenic emissions. This represents a significant advancement over previous attempts to generate limited temporal and spatial scope data (Gordon et al., 2018). These BC levels were among the highest in the world, exceeding BC concentrations reported for other polluted areas like China (Hang et al., 2023). Although we estimated high mortality risks attributable to BC exposure in this study, further epidemiological studies should be conducted to refine these estimates for the Indian population using primary health data. Secondly, the all-cause, CVD, and respiratory mortality burden attributable to BC in India was higher than the estimates for China (and for many other countries) because of higher BC exposure and higher MRs in India. Even if strategic actions reduce BC exposure in future, unless MRs (which also depend on other risk factors) are reduced through better healthcare management, the benefit of BC exposure reduction may not be fully realized in India. And finally, BC-attributable mortality per 100,000 population in India is higher than that attributable to ambient $\text{PM}_{2.5}$. Recent global studies, including a few from India, reported higher associations between several disease outcomes and exposure to BC compared to many other $\text{PM}_{2.5}$ species (Chaudhary et al., 2022, 2023; Park et al., 2018). This implies a higher toxicity of BC particles relative to $\text{PM}_{2.5}$, accounting for a mixture of species including BC. The World Health Organization (WHO) ascertained that BC should serve as an

indicator, along with $PM_{2.5}$, for the health impact assessments (Janssen et al., 2012). Our results suggest that reducing BC emissions can reduce ambient $PM_{2.5}$ by up to 15%, along with a larger health benefit per unit reduction.

Third, this study demonstrated substantial heterogeneity in BC exposure across demographic subgroups in India. Solid biomass combustion was identified as the dominant source of BC in rural areas, with additional seasonal contributions from open biomass burning, whereas urban BC sources were more spatially dispersed, resulting in greater within-urban exposure variability (Bond et al., 2013). Although absolute exposure disparities within subgroups declined consistently across the NFHS time points, relative disparities between subgroups remained largely unchanged (Table S5 in Supporting Information S1). The reduction in absolute disparity was primarily attributed to an overall decline in BC exposure nationwide, as supported by sensitivity analyses showing that exposure reductions alone would have lowered within-subgroup disparity by 28%–31% in both urban and rural regions. In contrast, minimal changes in across-subgroup disparity likely reflected a homogenization of exposure levels driven by reductions at pollution hotspots and/or increases in low-background areas. Despite improvements in overall air quality, relative BC exposure disparities persisted or increased across most subgroups, largely due to spatially heterogeneous sources and uneven benefits of mitigation measures (Katoch et al., 2023). Existing air pollution policies in India were predominantly urban-centric, city-wide, and sector-specific, and did not explicitly address intra-urban inequalities, resulting in disproportionate benefits to better-served areas while high-exposure clusters remained underserved (Bramble et al., 2023). These patterns highlighted the need to incorporate exposure disparities into targeted air quality mitigation strategies and to regularly update sector-specific emission inventories to track local- and regional-scale inequalities.

International experience further underscored the importance of equity-centered air quality governance. In the US, evidence of persistent exposure disparities informed the development of Environmental Justice Screening Tools (EJST) and regulatory frameworks aimed at monitoring and reducing inequalities across racial and income-based subgroups (Bramble et al., 2023; Jbaily et al., 2022). Similarly, China began integrating socioeconomic vulnerability indicators into urban air quality planning after documenting higher pollution exposure among disadvantaged sub-populations in rapidly urbanizing regions (Tu et al., 2022). Drawing on these examples, the study suggested that India could benefit from establishing a national air pollution disparity surveillance system to routinely monitor subgroup-level exposures and support adaptive, evidence-based policy interventions.

Limitations and Future research scopes: Our analysis has a few limitations that require discussion. *First*, our health impact assessment was based on the pooled relative risk estimates derived from global studies. Since there are not enough Indian cohorts, except a few such as HAPIN (Johnson et al., 2022) where BC personal exposure was measured, it is not feasible to assess the pooled estimate's representativeness fully. Future cohorts should systematically measure BC exposure to address this limitation, allowing for a more accurate estimation of the relationship between BC exposure and various health outcomes. *Secondly*, the limited availability of ground-level BC measurements remains a significant challenge. The sparse distribution of monitoring sites can introduce uncertainties in validating BC concentrations in areas and periods with insufficient data. This underscores the need for expanding BC monitoring networks across India, especially in underrepresented regions and, most importantly, making the data open for further utilization. *Third*, we assigned the same BC exposure to different subgroups residing within the buffer of each NFHS cluster. Such scale limitation may underestimate the disparity estimates, as a recent US-based study has reported increasing disparity in air pollutants with increasing spatial scale of exposure estimates (Johnson et al., 2022). Future research should formulate sub-km scale BC exposure and incorporate the updated demographic estimates from the census to improve the disparity assessments among the sub-populations. *Fourth*, we used the same exposure-response functions for the mortality assessments across the subgroups since subgroup-specific MRs are unavailable. This is a critical aspect and should be addressed in future GBD-India studies to identify the “true” vulnerability across the demographic subgroups for better policy. *Fifth*, future research should prioritize the expansion of regionally representative BC monitoring networks and the integration of post-pandemic observational data to further improve the accuracy and policy relevance of high-resolution BC exposure estimates, obtained in this study. And finally, our analysis focused specifically on assessing the spatio-temporal variability of BC exposure across India, quantifying its attributable mortality burden, and disparities across demographic subgroups. Therefore, evaluating alternative air quality or climate mitigation pathways such as sector-specific $PM_{2.5}$ reduction strategies, SO_2/NO_x controls, or technology-transition scenarios were beyond the scope of this study. Future studies should explicitly compare alternative

mitigation pathways using integrated assessment or scenario-based modeling frameworks, which would be valuable for prioritizing interventions across climate and public health objectives.

In conclusion, health is not an integral part of the current National Clean Air Program (NCAP) framework and existing socio-environmental policies (Gopikrishnan & Kuttippurath, 2024). Our findings highlighted BC as a critical contributor to ambient $PM_{2.5}$ -related health burden and exposure disparities in India, particularly among socioeconomically disadvantaged and rural sub-populations. Clean air actions are restricted locally, and the regional level action plans using an airshed approach are missing in India. If mitigation measures are prioritized in states showing higher (>0.7) BC-to- $PM_{2.5}$ all-cause mortality ratio (Figure S12 in Supporting Information S1), larger health benefits are expected. Note that India is yet to have a mitigation target for BC exposure, however, our health burden assessments depicted that an aggregated 1207446 all-cause premature deaths could be averted annually in India if BC exposure are completely mitigated; given its high toxicity per unit mass as compared to total $PM_{2.5}$ and its close association with combustion-related sources affecting vulnerable sub-populations. Additionally, in the absence of mitigation targets for BC exposure in India, we conducted sensitivity analyses where we estimate that 307881, 83662, and 32427 BC-attributable all-cause, CVD, and respiratory deaths could be avoided (Table S3 in Supporting Information S1), annually if the states reduce their mean BC exposure by 25%. Reducing BC exposure by 50%, an aggregated 644293, 174588, and 68023 all-cause, CVD, and respiratory mortalities could be prevented. With the strictest mitigation measures (that will reduce BC exposure by 75%), a cumulative 1012385, 273532, 107141 all-cause, CVD and respiratory deaths could be saved, respectively. Further, environmental equity is a new concept in India (de Souza et al., 2023), and current policies do not consider environmental justice as a criterion. Our results will help policymakers understand this issue. We advocate that policymakers implement targeted emission reduction strategies to protect the health of the vulnerable subgroups and achieve environmental and health equities in the foreseeable future. While reducing concentrations of $PM_{2.5}$ species that scatter solar radiation is expected to accelerate warming (Lelieveld et al., 2019), reducing BC concentrations will have the opposite effect because they absorb solar radiation. BC was found to weaken the Indian monsoon via radiative feedback (Meehl et al., 2012); hence, if reduced, it would strengthen the monsoon. Therefore, prioritizing BC reduction in the clean air action plan is the most strategic and effective ways to achieve climate and health co-benefits, and socio-environmental equities in India.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The raw data sets used to perform machine learning (ML) models and perform the BC exposure attributable health burden and disparity assessments are available at (Pratima & Sagnik, 2025).

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